



GeoDF-Adaptive (geometry-only, training-free, front-end module)

- (a)** Lift tracks to the normalized plane and map to pseudo-pixels
- (b)** Estimate temporal fundamental matrix F (RANSAC); score the Sampson residual e_i
- (c)** Dual gate: dynamic candidate iff (RANSAC outlier) AND ($e_i > \tau$)
- (d)** Scene-aware activation: arm only when EMA of outlier ratio $\geq p_{on} = \text{clamp}(\text{floor} \cdot 1.8 + 0.05, 0.10, 0.40)$
- (e)** Track-level temporal voting ($k = 2$, 30-frame warm-up) + ratio guard $\leq 40\%$ of features per frame
- (f)** Hard-delete confirmed dynamic tracks; keep the rest

Only the highlighted module is added; the stereo-inertial back-end is unmodified.